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AN EXPLORATORY STUDY OF THE BI-SPECTRUM OF ECONOMIC TIME SERIES

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1.0 Introduction

The main motivation of the analysis presented here is to investigate certain non-linear properties of the mechanism which may be supposed to generate a time series. I investigate specifically quadratic terms in the generating model. For a wide-sense stationary process the spectrum describes the linear mechanism. For a third moment stationary process the bi-spectrum describes the quadratic terms in the mechanism. I will discuss the derivation of the bi-spectrum. The interpretation of the bi-spectrum in terms of certain non-linear transformations will also be discussed.

As I have chosen to compute bi-spectra by means of complex-demodulation, there will be a major digression into the computation and analysis of complex-demodulates.

For background information concerning the estimation of spectra the reader is referred to Blackman and Tukey (1959). Analysis of certain non-linear systems is discussed in Wiener (1958), while the statistical theory of higher order spectra (polyspectra) is treated in Brillinger (1964). An extremely interesting application of the bi-spectrum is given in Hasselman *et al.* (1963).

2.0 Complex-Demodulation

While a variety of procedures for estimation of the bi-spectrum seem natural, the use of complex-demodulation recommends itself in terms of the generality of the procedure. In addition certain potentially useful by-products are produced. The procedure is general in the sense that, once the complex-demodulates have been computed, polyspectra of any order may be computed using any desired time interval within the realisation. The main by-products of this procedure for estimating bi-spectra are the complex-demodulates themselves.

2.1 Definition and Computation of Complex-Demodulates

Complex-demodulates are computed according to

$$x_{i,t} = L(x_t e^{i\omega_i t}), \quad \dots (2.1)$$

where $x_{i,t}$ is a *complex-valued* complex-demodulate, x_t is the original (real valued) time series, ω_i is the centre frequency for the demodulate, and $L(\)$ is a low-pass filter.

The form chosen for the low-pass filter may depend on a variety of considerations. A wide class of filters is defined by a generating function of the form:

$$A(z) = \frac{\sum_{i=-k}^l a_i z^i}{1 - \sum_{i=1}^m b_i z^i} \quad \dots (2.2)$$

using the z -transform given by

$$u(z) = \sum_{t=-\infty}^{\infty} u_t z^t$$

and where a_i , b_i are constants, and $A(z)$ is the filter function, which for $z = e^{i\omega}$ gives the frequency response function of the filter.

Equation (2.2) may be written as a linear regressive process of the form

$$y_t = \sum_{i=-k}^l a_i x_{t-i} + \sum_{i=1}^m b_i y_{t-i} \quad \dots (2.3)$$

If the complex-demodulates are to be used for prediction purposes it might be reasonable to set $k = -1$ and estimate the a_i and b_i so as to minimise the mean square error of y_t taken as an estimate of x_t .

On the other hand, the complex-demodulates may be computed in order to form an historical description and possibly, to estimate spectra. In this case it would seem more natural to use a filter which results in a symmetric moving average operator.

From the standpoint of computational speed a very great advantage can be realised through the use of filter functions which are formed by powers of filter functions of the form

$$K(\omega) = \frac{\sin m\omega}{\sin \omega} \quad \dots (2.4)$$

The value of filter functions of this form in terms of computational speed is that they imply a rectangular data window of the form

$$y_t = \sum_{i=-m}^m x_{t-i} \quad \dots (2.5)$$

We may obtain powers of equation (2.4) by iteration of equation (2.5) on the data.¹ Thus, instead of having to employ a data window of the form:

$$y_t = \sum_{i=-m}^m a_i x_{t-i} \quad \dots (2.6)$$

we may convolute equation (2.5) r times in order to obtain a filter with

¹ Prof. John Tukey first pointed out to me the advantages of this procedure in terms of numerical method.

transfer function which is the r th power of $K(\omega)$. As is well known $r = 2$ gives the Bartlett window

$$K_2(\omega) = \left(\frac{\sin m\omega}{\sin \omega} \right)^2, \quad \dots (2.7)$$

while $r = 4$ gives the Parzen window

$$K_4(\omega) = \left(\frac{\sin m\omega}{\sin \omega} \right)^4. \quad \dots (2.8)$$

For the computations in this paper we iterated equation (2.5) four times with a constant value of m in order to produce a Parzen window of $4m - 3$ terms. A direct application of the data window corresponding to a Parzen kernel would have required approximately

$$\{N - (4m - 3)\} (4m - 3) / 2 + 1 \quad \text{multiplications}$$

$$\text{and} \quad \{N - (4m - 3)\} (4m - 3) \quad \text{additions}$$

where N is the number of observations. The iteration of equation 2.5 requires only

$$\sum_{i=1}^4 \{(2m + 1) + 2(N - 2im)\} \quad \text{additions}$$

and no multiplications. For the series used in this paper the 4-fold iteration was nearly ten times faster than the direct weighted moving average when run on an IBM 7094.

2.2 Analysis of Complex-Demodulates

Complex-demodulates may be viewed as time series which estimate the variance of a time series within a frequency band.

The time average over the interval t_1 to t_2 of the modulus squared of the complex-demodulate at frequency ω_0 estimates the spectral density in the frequency band $\omega_0 \pm \Delta\omega$ from the realisation from t_1 to t_2 . $\Delta\omega$ is given by the bandwidth of the low-pass filter.

The real and imaginary parts of the complex-demodulates describe the variation of the series within a frequency band, $\omega_0 \pm \Delta\omega$. The phase of a demodulate gives the relative phase of the variation in the band $\omega_0 \pm \Delta\omega$ as a function of time. Thus the demodulates may be used to investigate evidence of non-stationarity of the time series within each frequency band. The series shown in Fig. 2.1 are interesting in this respect. At $t \approx 24$ there is apparently a considerable change in the behaviour of some of the demodulates. Estimates of the spectra before and after this time point also indicate that the power at certain frequencies changed considerably. The point of interest in terms of the economic analysis of this series is that a major institutional change occurred at exactly this point in time. This point will be discussed in more detail in Section 5.0. We may simply observe at this point that this institutional change was reflected in changes in the spectrum at some frequencies but not at others.

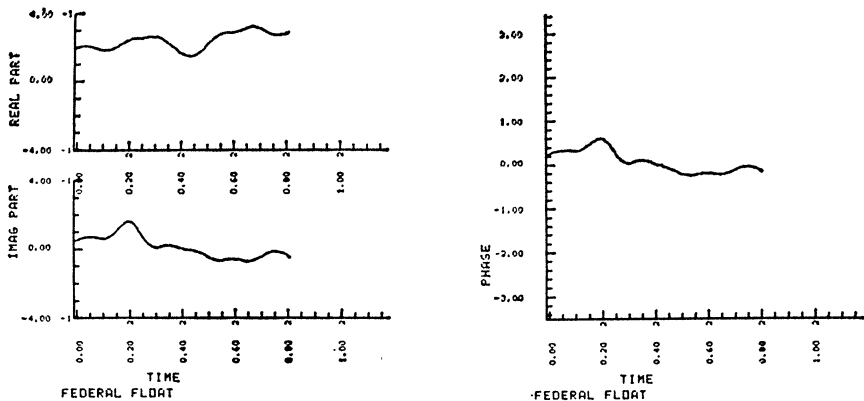


FIG. 2.1.1 frequency $\pi/6$

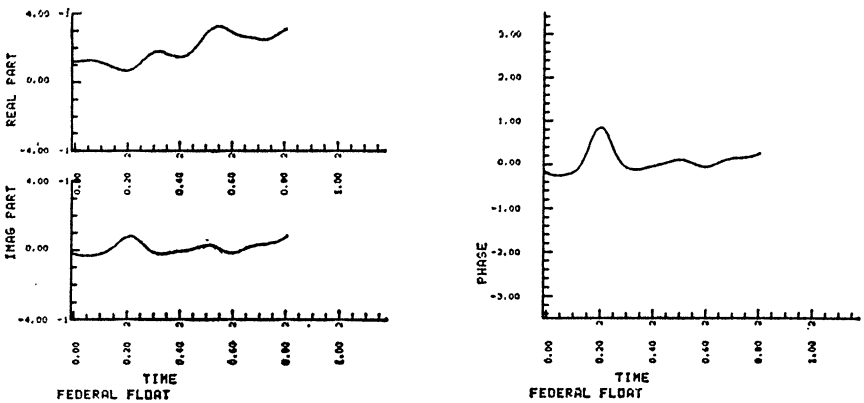


FIG. 2.1.2 frequency $\pi/3$

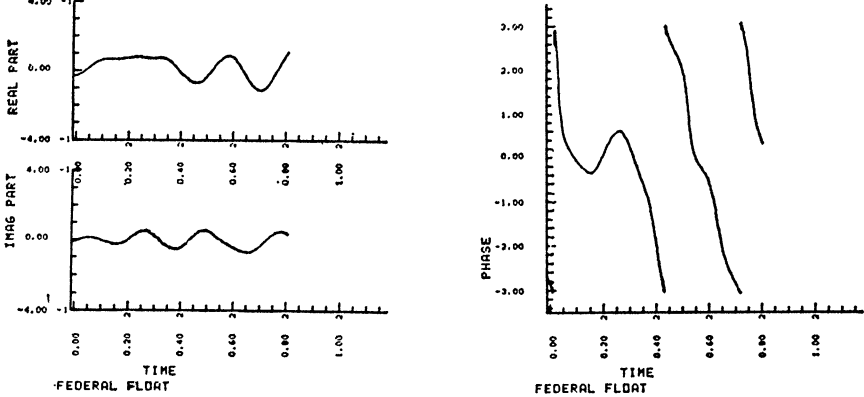


FIG. 2.1.3 frequency $11\pi/24$

FIG. 2.1. Complex-Demodulates of Monthly Federal Float data. 227 observations, 15 terms per convolution, 4 convolutions. Demodulates decimated by 2.

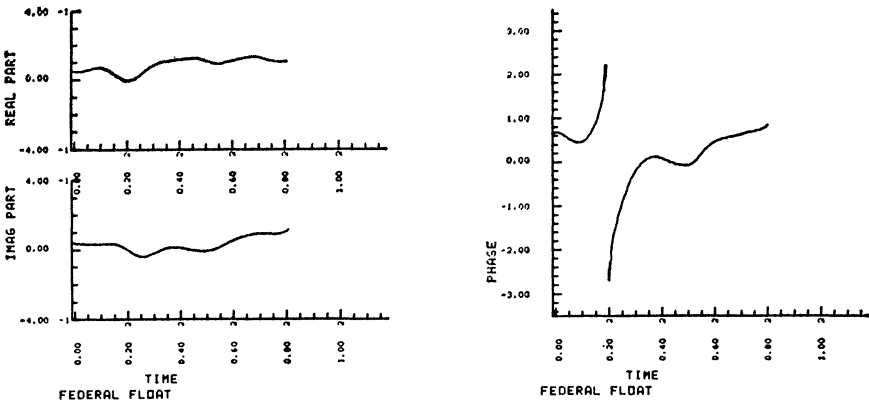


FIG. 2.1.4 frequency $\pi/2$

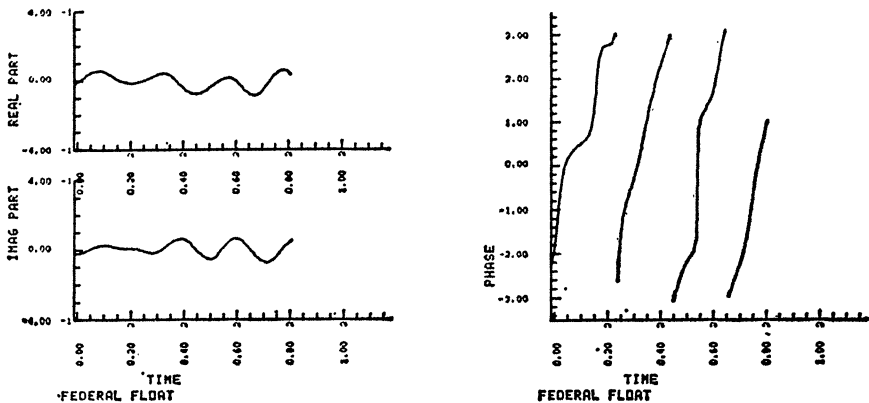


FIG. 2.1.5 frequency $13 \pi/24$

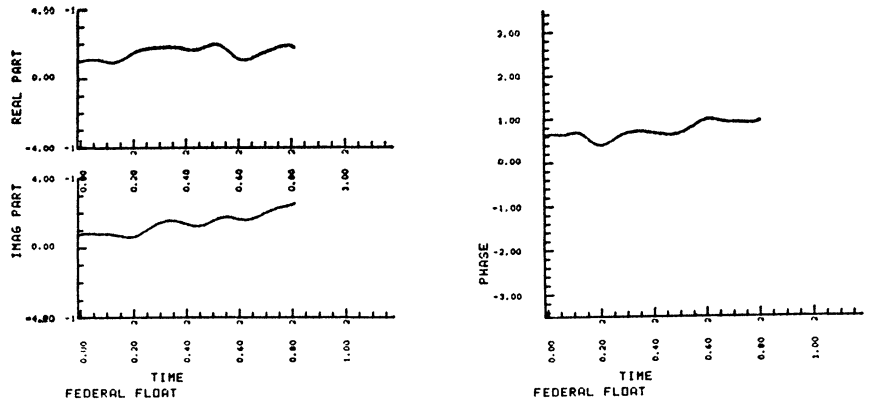


FIG. 2.1.6 frequency $2 \pi/3$

FIG. 2.1. Complex-Demodulates of Monthly Federal Float data. 227 observations, 15 terms per convolution, 4 convolutions. Demodulates decimated by 2.

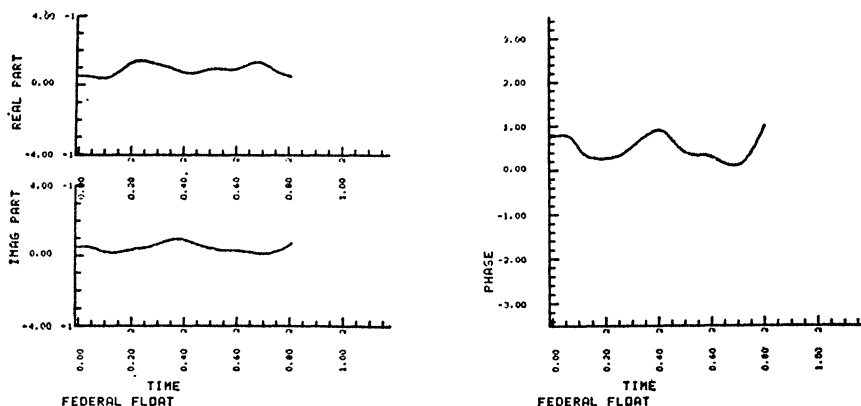
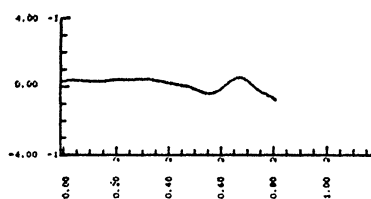
FIG. 2.1.7 frequency $5\pi/6$ FIG. 2.1.8 frequency π

FIG. 2.1. Complex-Demodulates of Monthly Federal Float data. 227 observations, 15 terms per convolution, 4 convolutions. Demodulates decimated by 2.

Fig. 2.1 also shows another use of demodulates. The phase of a demodulate has a well defined behaviour in the vicinity of a peak in the spectrum. To see the effect of a peak in the spectrum at ω_1 we may consider a series

$$x_t = ae^{i\omega_1 t} \quad \dots (2.9)$$

If we demodulate at ω_0 and filter with an ideal low-pass filter with bandwidth greater than $|\omega_1 - \omega_0|$ we then have

$$x_{0,t} = ae^{i(\omega_1 - \omega_0)t} \quad \dots (2.10)$$

If the phase of the demodulate is given by

$$\Phi_{0,t} = \tan^{-1} \frac{\text{Im} [x_{0,t}]}{\text{Re} [x_{0,t}]} \quad \dots (2.11)$$

then we have

$$\Phi_{0,t} = (\omega_1 - \omega_0)t \quad \dots (2.12)$$

We may use equation (2.12) to help to locate the peak of a spectrum and to study the time stability of the peak frequency. The three phase diagrams in Fig. 2.1 show the effect of demodulating just below, exactly on, and just above a peak in the spectrum. The phase diagram

D*

for the demodulate at $\omega_0 = \pi/6$ shows that the peak is quite stable in terms of frequency. However, the amplitude of the demodulate varies over a considerable range.

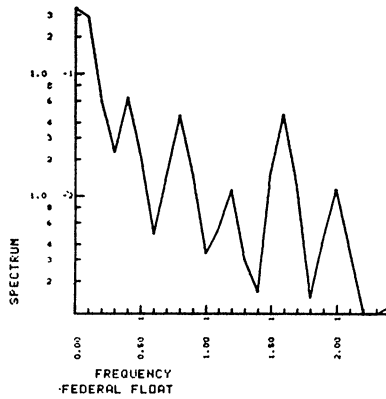


FIG. 2.2. Spectrum of Monthly Federal Float data computed from complex-demodulates shown in Fig. 2.1.

The peaks in the spectrum of this series, shown in Fig. 2.2, correspond to the seasonal frequencies. The complex-demodulates at these frequencies may be used in an attempt to estimate the variation in the series attributable to the seasonal periodicity. The complex-demodulates indicate an important aspect of the seasonal adjustment problem in that, while the frequency is very nearly constant, the amplitude of each of the seasonal frequencies seems to vary quite rapidly. As a result there are also changes in the phase.

2.3 *Correlation between Complex-Demodulates*

On the basis of linear theory the expected correlation between two complex-demodulates is given by the overlap of the frequency bands associated with the demodulates. This is directly analogous to the correlation between estimates of the spectrum. We can calculate the expected correlation in the following manner. If the low-pass filter used is given by $L(\omega)$, we may introduce the window $L(\omega - \omega_i)$ which is the filter $L(\omega)$ centred at ω_i . The expected coherence between complex-demodulates at ω_i and ω_j is effectively zero if the bandwidth of $L(\omega - \omega_i)$ is small when compared to $\omega_i - \omega_j$.

If, in choosing the set of frequencies at which to demodulate, we follow a practice analogous to that of estimating the spectrum of a set of equi-spaced points equal to the number of lags used for the correlation coefficient, then for a low-pass filter of bandwidth comparable to a Bartlett or Tukey window we would expect all of the complex-demodulates to be nearly independent. If we use a 'Parzen-like' window for the low-pass filter then we would expect some correlation between adjacent demodulates. As we have chosen a filter with a bandwidth which is very narrow when compared to the distance between

the frequencies at which we demodulated we would expect very low coherence on the basis of linear theory.

In order to investigate the validity of the assumption of linear theory in this respect we may consider computing the cross-spectra between complex-demodulates. Consideration of the mechanism which generated our data led us to suppose that there might be some correlation between demodulates from the seasonal frequencies. In order to investigate this possibility we computed cross spectra of the absolute magnitudes of the complex demodulates of the monthly 'Federal Float' data for each of the pairs of seasonal frequencies. The average coherences in the frequency range 0 to .5 cycles per year are shown in Fig. 2.3. Only the average values are given since in most cases the coherence did not change greatly in this range.

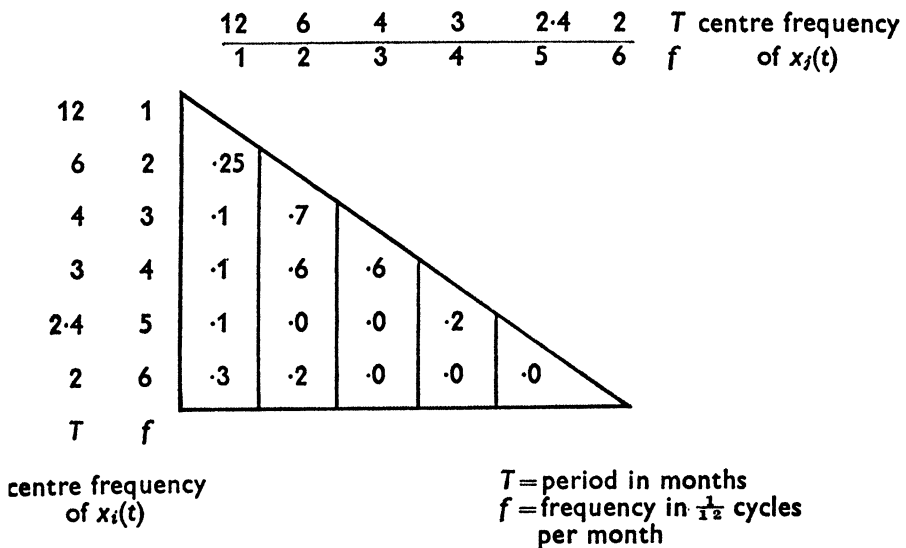


FIG. 2.3. Average value of coherence between complex-demodulates.

It is evident that the frequency pairs (3, 2), (4, 2), and (4, 3) seem to be coupled in a non-linear way.

3.0 The Bi-spectrum

The bi-spectrum is the frequency decomposition of the third moment of a stochastic process. We will first discuss the derivation of the bi-spectrum and other related quantities. We will then discuss the estimation of bi-spectra. Finally we will present the results of estimation using economic time series.

3.1 Derivation of the Bi-spectrum and Related Quantities

The bi-spectrum of a multivariate time series $\{X_1(t), X_2(t), X_3(t)\}$ may be defined as the third order Fourier transform of the third moment of the multivariate process:

$$f_{123}(\omega_1, \omega_2, \omega_3) = \frac{\iiint \gamma_{123}(\tau_1, \tau_2, \tau_3) \exp\{i(\omega_1\tau_1 + \omega_2\tau_2 + \omega_3\tau_3)\} d\tau_1 d\tau_2 d\tau_3}{\delta(\omega_1 + \omega_2 + \omega_3)}, \dots (3.1)$$

where $f_{123}(\omega_1, \omega_2, \omega_3)$ is the bi-spectrum given for

$$\omega_1 + \omega_2 + \omega_3 = 0$$

$\gamma_{123}(\tau_1, \tau_2, \tau_3)$ is the third order product moment, or cumulant for $X_1(\tau_1)$, $X_2(\tau_2)$, and $X_3(\tau_3)$, and $\delta(\omega_1 + \omega_2 + \omega_3)$ is the Dirac delta function.

This notation, while not so obviously related to the usual notation for the spectrum, has the advantage that it maintains symmetry in the three variables and will be convenient in the ensuing presentation. We could have introduced the constraint that $\omega_1 + \omega_2 + \omega_3 = 0$ through the use of only two frequencies (ω_1, ω_2) and two lag variables (τ_1, τ_2) . While conventional notation indicates the spectrum as a function of one frequency variable, it seems preferable to think of the spectrum as a function of two frequency variables which sum to zero in order to relate it to the bi-spectrum.

We now make the assumption that the process is stationary to the third order in order to justify our estimates of the bi-spectrum which rely on time-averages. We might consider estimation of the bi-spectrum through the following steps. First, we would compute the lagged product moment

$$c_{123}(\tau_1, \tau_2, \tau_3) = \frac{1}{N} \sum_{t=1-\min(\tau_1, \tau_2, \tau_3)}^{N-\max(\tau_1, \tau_2, \tau_3)} X_1(t-\tau_1) X_2(t-\tau_2) X_3(t-\tau_3) \dots (3.2)$$

$$\tau_1 = 0, \pm 1, \dots \pm m$$

$$\tau_2 = 0, \pm 1, \dots \pm m$$

$$\tau_3 = 0$$

Since the product moment is invariant to a translation in time we may 'centre' the estimate on $\tau_3 = 0$.

We might either apply a lag window to $c_{123}(\tau_1, \tau_2, \tau_3)$ or use a window in the Fourier transformation of $c_{123}(\tau_1, \tau_2, \tau_3)$. In the later case we have

$$\hat{f}_{123}(\omega_1, \omega_2, \omega_3) = \sum_{\tau_1=-m}^m \sum_{\tau_2=-m}^m K(\tau_1, \tau_2) c_{123}(\tau_1, \tau_2, \tau_3) \exp\{i(\omega_1\tau_1 + \omega_2\tau_2)\}, \dots (3.3)$$

where $K(\tau_1, \tau_2)$ is a two dimensional lag window whose Fourier transform is such that neighbouring frequencies are averaged in order to yield estimates of the bi-spectrum which have desirable sampling properties.

While the approach indicated above is, again, a natural extension of the estimation procedures used for the spectrum we have adopted an

alternative approach in practice which derives directly from our interest in complex demodulates. Given that we have computed complex-demodulates as indicated in Section 2, we form an estimate of the bi-spectrum simply by taking sums of triple products of the demodulates. Thus we compute

$$\hat{f}_{123}(\omega_1, \omega_2, \omega_3) = \frac{1}{N'} \sum_{t=1}^{N'} X_{1,\omega_1}(t) X_{2,\omega_2}(t) X_{3,\omega_3}(t), \quad \dots (3.4)$$

subject to $\omega_1 + \omega_2 + \omega_3 = 0$, where N' is the number of available values of the complex-demodulates, $X_{1,\omega_1}(t)$ is the complex-demodulate of $X_1(t)$ at ω_1 , and $\hat{f}_{123}(\omega_1, \omega_2, \omega_3)$ is the estimate of the bi-spectrum.

This procedure, together with our computation of complex-demodulates provides us with a computationally efficient and flexible means of estimation for the bi-spectrum. Among the advantages of this procedure are the facility with which we can compute a series of bi-spectra over different time intervals without recomputation of the demodulates and the opportunity to compute the spectrum at the same time that we compute the bi-spectrum with almost zero cost. We compute the spectrum simply by forming the sum

$$\hat{f}_{12}(\omega_1, \omega_2) = \frac{1}{N'} \sum_{t=1}^{N'} X_{1,\omega_1}(t) X_{2,\omega_2}(t), \quad \dots (3.5)$$

subject to $\omega_1 + \omega_2 = 0$, where $\hat{f}_{12}(\omega_1, \omega_2)$ is an estimate of the spectrum.

If we wish to compute the bi-spectrum of a multivariate process involving three distinct series $X_1(t)$, $X_2(t)$, $X_3(t)$ the bi-spectrum is a complex valued function defined on a square two dimensional region in the frequency plane centred on $\omega_1 = \omega_2 = \omega_3 = 0$ with sides equal to 2π . If either two or all three of the series are identical then symmetry properties permit reduction of the area within which we must compute estimates. In particular if

$$X_1(t) = X_2(t) = X_3(t)$$

then we need only compute estimates in the region

$$0 \leq \omega_1 \leq \pi$$

$$0 \leq \omega_2 \leq \frac{\pi}{2}$$

$$0 \leq \omega_1 + 2\omega_2 \leq \pi$$

However, the bi-spectrum is still complex-valued. Thus, we note that while the spectrum of a single real-valued time series is real-valued the bi-spectrum of a single time series is complex-valued.

Before we discuss the empirical results of our exploratory analysis we must present some functions of the bi-spectrum. These functions will be found useful when we consider certain hypotheses about the generating process for our data.

4.0 Functions of the Bi-spectrum

While the bi-spectrum alone contains the information about the third moment which one wants to study, this information may be more easily interpreted if certain functions of the bi-spectrum are computed. The functions of the bi-spectrum which may be interesting will depend on specific information or hypotheses about the nature of the generating process.

One basic sort of measure which we may want to consider is how much of the variation in the series comes about from products of pairs of series. In other words, how well does a model of the form

$$X_k(t) = \beta X_i(t) X_j(t) \quad \dots (4.1)$$

represent the process? This question leads us to estimate the bi-spectral coefficient

$$\hat{S}_3(\omega_1, \omega_2, \omega_3) = \frac{f_{123}(\omega_1, \omega_2, \omega_3)}{\{f_1(\omega_1)f_2(\omega_2)f_3(\omega_3)\}^{\frac{1}{4}}}, \quad \dots (4.2)$$

where $f_i(\omega_i)$ is an estimate of the spectrum of $X_i(t)$ at ω_i .

The estimate of the bi-spectral coefficients provides us with an estimate which should not have peaks which are due to power in the spectra.

We may think of the process which generates the series at being capable of representation in terms of a power series expansion. We then have

$$X_k(t) = \sum \alpha_i X_i(t) + \sum \beta_i X_i X_j + \dots \quad \dots (4.3)$$

In order to estimate the quadratic transfer function in the frequency domain, we will consider estimates of this transfer function of the form

$$\hat{T}_3(\omega_1, \omega_2 | \omega_3) = \frac{f_{123}(\omega_1, \omega_2, \omega_3)}{f_1(\omega_1)f_2(\omega_2)} \quad \dots (4.4)$$

where $\hat{T}_3(\omega_1, \omega_2 | \omega_3)$ is an estimate of the quadratic transfer function of products of $X_1(t)$ and $X_2(t)$ producing power in $X_3(t)$.

We may also consider the problem of whether some transformation of a series will result in a 'more nearly linear' generating process. If we represent the series in terms of an expansion of the form

$$X_1(t) = Y(t) + \alpha Y^2(t) \quad \dots (4.5)$$

we will want to consider the following estimate in order to determine whether a transformation such as this exists. The estimate is given by

$$\hat{P}_3(\omega_1, \omega_2, \omega_3) = \frac{f_{123}(\omega_1, \omega_2, \omega_3)}{2\{f(\omega_1)f(\omega_2) + f(\omega_1)f(\omega_3) + f(\omega_2)f(\omega_3)\}} \quad \dots (4.6)$$

In particular, for the data that we have studied we wanted to investigate the question as to whether a logarithmic transformation would lead to a more nearly Gaussian process. If the transformation

$$Y(t) = \log X(t) \quad \dots (4.7)$$

would produce a process for the $X(t)$ which was Gaussian then we would expect

$$\hat{P}_3(\omega_1, \omega_2, \omega_3) = -\frac{1}{2}. \quad \dots (4.8)$$

If, on the other hand, $X(t)$ were linear then

$$E(\hat{P}_3(\omega_1, \omega_2, \omega_3)) = 0. \quad \dots (4.9)$$

5.0 Analysis of Data

Two kinds of time series were analysed in this exploratory study. The first of these were series of stock market price data from the New York stock exchange. The other series was a series of monthly observations of one of the monetary variables of the U.S. Federal Reserve System. The reasons for choosing the first of these two series as an example were that the series was 'simple' in the sense that the spectrum of the first differences of the series was known to be effectively flat, and it was not expected that the bi-spectrum would be very complicated. However, the question of whether a logarithmic transformation would yield a simpler series was one of some interest. The second series, on the other hand, was chosen because of a pronounced seasonal variation and because it represented a relatively long and, except for one known change in the process, apparently fairly homogeneous series.

5.1 Results of Analysis of Stock Market Series

For the purposes of this exploratory study the stock market data were used simply to investigate the question of the effect of a logarithmic transformation. We attempted to answer two specific questions:

1. Will a logarithmic transformation uniformly reduce the process to a linear, second-order process, or will there remain power in the bi-spectrum at certain frequencies which evidently needs to be explained by a non-linear mechanism?
2. Should the mean of the series be partially removed before applying the logarithmic transformation? In other words, is the coefficient a in an equation of the form

$$y_t = \log(x_t + a) \quad \dots (5.1)$$

equal to zero or does the transformation 'work better' if a negative constant is inserted before taking logarithms?

The results of two of the computations are shown in Figs. 5.1 and 5.2. Fig. 5.1 shows the function

$$\hat{P}(\omega_1, \omega_2, \omega_3) = \frac{f_{123}(\omega_1, \omega_2, \omega_3)}{2\{f(\omega_1)f(\omega_2) + f(\omega_1)f(\omega_3) + f(\omega_2)f(\omega_3)\}} \quad \dots (5.2)$$

for the time series of transactions prices on the New York stock exchange for International Telephone and Telegraph (ITT) for the month of April 1963. This series is made up of the price of each transaction in

	ω_2												$\pi/2$
	1	2	3	4	5	6	7	8	9	10	11	12	
0	1.8931												
1	0.1783	-0.0335											
2	0.8783	0.4889	0.4624										
3	0.9463	0.1418	0.5233	0.6564									
4	0.3308	0.4341	0.3459	-0.1430	-0.3275								
5	0.9444	0.3139	-0.2925	0.0091	0.4927	-0.2465							
6	0.9598	0.1505	-0.0141	0.4907	0.4325	-0.0978	0.3154						
7	0.0822	0.1515	0.5604	0.4308	0.0555	-0.2979	0.0336	0.0010					
8	0.6188	0.0804	-0.0079	0.3409	0.4707	-0.2769	0.1498	0.3836	0.1485				
9	0.1126	0.0685	0.1928	-0.0120	0.0466	-0.1107	0.0377	0.1483	0.1785	0.2920			
10	0.0690	0.1684	0.0562	-0.0703	0.3766	0.4829	-0.1939	0.1977	-0.0709	0.0544	-0.3089		
11	0.0757	-0.1783	-0.2067	0.0099	0.0596	-0.1939	0.2865	0.1677	-0.0103	-0.2393	-0.5376	-0.2748	
12	0.1560	0.1392	-0.3295	0.4098	0.1563	0.0468	-0.1300	-0.1677	0.1588	-0.1831	-0.0930		
13	-0.1333	0.0348	0.1354	-0.1086	0.0685	-0.1719	-0.0353	0.1057	-0.1923	-0.0878			
14	0.0349	-0.1367	0.0683	0.1205	0.2068	-0.0235	-0.0874	-0.2903	-0.0051	0.2009			
15	-0.0432	0.0796	-0.0600	0.4556	0.3003	0.1285	0.0108	-0.0051	-0.0960				
16	0.0853	-0.2120	-0.3745	-0.1042	0.0944	-0.0865	0.1355	-0.0960					
17	0.7097	-0.0643	-0.1342	-0.2051	-0.2963	-0.2041	-0.6031						

Fig. 5.1 $\hat{P}(\omega_1, \omega_2, \omega_3)$
Price Series for International Telephone and Telegraph
890 observations (individual transactions, extend over about a month)
25 terms in each of four iterations of filter on complex-demodulates
First differences of original price data.

ω_1	0	1	2	3	4	5	6	7	8	9	10	11	12
	2.0573	0.0516											$\pi/2$
1	0.2555	0.5614	0.5206										
2	1.0096	0.1985	0.5796	0.6971									
3	1.0173	0.4816	0.3593	-0.1511	-0.3473								
4	0.4367	0.3483	-0.2777	-0.1067	-0.0119	0.5034							
5	1.0089	0.1480	0.0349	0.5022	0.4341	-0.0976	-0.2409						
6	1.0494	0.1802	0.5892	0.4258	0.0377	-0.0691	-0.2954	0.3127					
7	0.1040	0.1080	0.0033	0.3203	0.4856	-0.2427	-0.2303	0.0145	-0.0026				
8	0.7138	0.0712	0.1942	-0.0144	-0.0435	0.0847	-0.1166	0.1314	0.4003	0.1410			
9	0.1696	0.1879	0.0616	-0.0777	0.3673	0.4954	0.0391	-0.0573	0.1475	0.1706	0.2978		
10	0.0967	-0.1565	-0.2126	-0.0046	0.0462	-0.0313	-0.1883	0.2869	0.1845	-0.0751	0.0536	-0.3056	
11	0.1169	0.1552	-0.3112	0.4162	0.1470	-0.6510	0.0642	-0.1270	-0.1635	-0.0150	-0.2326	-0.5338	-0.2792
12	0.1993	0.0487	0.1424	-0.1256	0.0604	-0.0067	-0.1682	-0.0402	0.1024	0.1621	-0.1893	-0.1011	
13	-0.0834	-0.1125	0.0793	0.1122	0.1645	0.2157	-0.0158	-0.0845	-0.2691	-0.1819	-0.0821		
14	0.0669	0.0762	-0.0663	0.4662	0.2834	0.3018	0.1388	0.0215	0.0053	0.2014			
15	-0.0185	-0.2157	-0.3793	-0.1019	-0.0603	0.1046	-0.0791	0.1366	-0.0895				
16	0.1243	-0.0486	-0.1323	-0.2051	-0.0830	-0.2912	-0.2181	-0.6252					
17	0.7523	-0.1944	-0.1819	0.1958	-0.0295	0.0089	0.1987						
18	0.2694	0.1074	0.1917	0.0210	0.0188	0.0182							
19	0.1510	0.2194	0.0800	0.1030	0.1753								
20	0.1262	0.2427	-0.1784	0.2611									
21	0.2350	-0.0689	-0.0657										
22	0.1232	0.1335											
23	-0.0357												
24	0.3409												
π													

Fig. 5.2 $\hat{P}(\omega_1, \omega_2, \omega_3)$
Price Series for International Telephone and Telegraph
890 observations
25 terms in each of four iterations of filter on complex-demodulates
First differences of natural logarithms of original data.

ITT which was recorded during this period. The number of transactions was 890.

We may postulate a model

$$X(t) = e^{Y(t)} \quad \dots (5.3)$$

where $X(t)$ is the observed series, and $Y(t)$ is a series from an underlying linear process.

If we expand $e^{Y(t)}$ in a power series about a point k we have

$$X(t) = e^k [1 + \{Y(t) - k\} + \frac{1}{2}\{Y(t) - k\}^2 + \frac{1}{6}\{Y(t) - k\}^3 \dots] \quad \dots (5.4)$$

Since k in some sense determines the 'origin' of the process we would expect it to be relatively small. However, quite surprisingly we find that the estimates derived from the transactions data are nearly invariant to the logarithmic transformation. Fig. 5.1 shows the estimate for the original data while Fig. 5.2 shows the estimate of natural logarithms of $X(t)$.

If, on the other hand, we compute these same estimates using data which extend over a longer time period and for which the sampling interval is much longer we find that the estimates are noticeably changed by taking logarithms. Figs. 5.3 and 5.4 give the estimates without and with the logarithmic transformation for the Standard and Poor monthly stock price index from 1871 to 1956. In addition, Fig. 5.5 shows the result of transforming the data by

$$X'(t) = X(t) - a \quad \dots (5.5)$$

where $a = \min [X(t)] + 1$.

From the estimate shown in Fig. 5.5 it seems that the process is such that the value of the coefficient a in equation (5.5) which makes the process appear to be nearly linear is close to zero.

From these estimates we can conclude that at the lower frequencies (i.e. less than about $\frac{1}{2}$ cycle per week) the logarithmically transformed process appears nearly linear; however, at higher frequencies the logarithmic transformation has very little effect. We would not expect the transformation to have much effect at high frequencies due to the fact that the variance at these frequencies is small. Thus, these results support the hypothesis of a log-linear process.

6. Analysis of Federal Reserve Data

The time series analysed in this section was made up of monthly observations of the quantity of cash held within the Federal Reserve System for the purpose of meeting inter-bank check handling obligations. Thus the variable being measured is the amount of funds required for the internal operations of the Federal Reserve with respect to the processing of checks. The higher the level of activity in the Federal Reserve banks the greater the demand for the funds in this 'Federal Float'. Therefore, this fund may be used to measure the level of activity of the member banks in the Federal Reserve System. Because

ω_1	1	2	3	4	5	6	7	8	9	10	11	12
0	-2.1051											
1	-1.0822	0.1294										
2	-0.8400	-0.2095	-0.1217									
3	-0.8155	-0.4434	-0.4928									
4	-1.8700	-0.4678	-0.7939	-0.5967								
5	-1.7769	0.1317	-0.4728	-1.3306	1.3128							
6	-0.8656	0.1208	-0.9413	-0.5582	-0.8291	-0.7599						
7	-1.1529	-0.8094	-1.3349	0.3627	-1.4988	-1.5827	-1.7480					
8	-1.6935	-0.9675	0.3791	-1.3199	-1.4948	-1.0272	-1.1255	-0.4502				
9	-0.9970	0.1368	-0.1977	-0.8395	-1.2933	-0.6226	-0.6431	-1.1495	-1.0627			
10	-1.2961	-0.7478	-0.4650	0.2689	0.6288	0.3524	0.6841	0.0383	-0.6771	0.3030		
11	-1.7655	0.5547	-0.8285	-1.5231	-0.1664	-0.9827	-1.0305	-0.7661	-0.5548	1.4858	-0.3104	
12	-0.5572	-0.5099	-0.7131	-0.3804	-1.1200	-1.0276	-1.5200	-0.8432	-1.4133	0.8215	-0.7668	-0.8134
13	-1.8102	-0.7491	-1.3881	-1.0308	-1.0695	-0.8975	-1.1868	-1.8800	-1.2921	0.5633	0.1543	
14	-1.1201	0.0961	-0.4842	-1.2470	-0.2006	-0.3221	-1.8469	-1.5141	-0.8446	-0.8149		
15	-1.7286	-0.3052	-1.6168	-1.0958	-0.1449	-1.6903	-1.7130	-1.3241	0.8534			
16	-1.0601	-0.8220	-0.3436	-0.5496	-0.6129	-0.8084	-1.0285	0.2080				
17	-1.6228	-0.1367	0.3228	-0.0348	-1.7193	-1.4925	1.2440					
18	-0.7066	0.4226	-0.1318	-2.1850	-1.7193	0.7039						
19	-1.2566	0.1081	-1.2909	-1.8172	-1.2445							
20	-0.5132	-0.8802	-1.1612	-0.5055	-0.0313							
21	-2.3368	-0.4803	-1.2536	-0.5439								
22	-1.0261	0.0870	0.7395	0.1214								
23	-0.9396	0.3723										
π	-1.5131											

FIG. 5.3 $\hat{P}(\omega_1, \omega_2, \omega_3)$

Based on first differences of Standard and Poor stock Price Index
(1020 weekly observations;

25 terms in each of four iterations of filter on complex-demodulates)

ω_1	1	2	3	4	5	6	7	8	9	10	11	12
0	-1.3198											
1	-0.6843	-0.0219										
2	-0.4836	-0.3700	-0.1164									
3	-0.5531	-0.0822	0.0255	-0.1361								
4	-0.6351	0.4174	0.5706	-0.2089	-0.0669							
5	-0.8870	0.5070	0.4761	0.2068	0.6952	1.4566						
6	-0.4815	0.0337	-0.6674	-0.1350	1.0909	0.7768	0.2313					
7	-0.6538	-0.6153	-0.8527	0.2147	0.7089	0.2560	0.0864	0.2179				
8	-1.4456	-0.4981	0.3217	0.1717	-0.3276	-0.6500	0.1634	0.5641	0.3871			
9	-0.5839	0.3805	0.3247	0.0760	-0.1053	-0.2858	0.7630	0.7115	0.0210	-0.4346		
10	-0.8392	0.0135	0.2014	-0.0791	-0.2294	0.9458	1.0326	0.6131	0.2434	0.0915	0.1582	
11	-1.2144	0.4479	0.1018	-0.0080	-0.2801	1.2345	0.4721	0.0550	0.1230	-0.1398	1.5506	0.9373
12	-0.3502	0.0762	0.3059	0.2125	0.4681	0.4404	0.1811	0.1801	0.0522	0.0740	0.4418	-0.5317
13	-0.7577	-0.2478	-0.7157	0.0175	-0.0219	-0.2173	-0.2858	-0.2759	-0.8627	-0.0689	-0.1275	-0.0320
14	-0.8505	-0.0643	-0.4130	0.2380	0.1874	0.4972	0.2058	-0.6756	-0.3236	-0.2425	-0.7325	
15	-1.3084	-0.5069	-1.0624	0.0364	0.5802	0.8179	-0.8171	-0.6862	-1.1256	-0.5819		
16	-0.9753	-0.3657	0.1485	0.5165	0.0304	0.9027	0.2225	-0.8699	-0.7738			
17	-1.1525	0.0907	0.1829	-0.0653	0.3101	0.3734	-0.9781	-0.5432				
18	-0.4305	-0.0303	-0.0644	-0.1749	-0.3818	-0.4065	-0.1828					
19	-0.6692	0.1267	-0.9295	-0.3607	-0.4082	-0.7587						
20	-0.1462	-0.5500	-0.8425	-0.4917	-0.7992							
21	-1.5379	-0.0953	-1.0988	-0.5141								
22	-0.2545	-0.2797	-0.1579									
23	-1.0322	-0.4237										
24	-1.5737											

FIG. 5.4 $\hat{P}(\omega_1, \omega_2, \omega_3)$

Based on first differences of Standard and Poor stock Price Index
(1020 weekly observations;
25 terms in each of four iterations of filter on complex-demodulates)

π	ω_1	1	2	3	4	5	6	7	8	9	10	11	$\pi/2$
0		-0.3328											12
1		-0.5422	-0.5634										
2		-0.4876	-0.5909										
3		-0.5622	-0.4381	0.0831									
4		-0.3248	-0.3685	-0.0584	0.1905								
5		-0.3737	0.0648	0.5271	0.1982	0.3625							
6		-0.3212	-0.4398	-0.3285	0.5657	0.9827	1.2527						
7		-0.5125	-0.7327	-0.2228	-0.3285	0.7093	1.0879	0.7093					
8		-1.1068	-0.3432	0.5397	0.6510	0.9575	1.0496	0.9575	1.6797				
9		-0.4207	0.2843	0.4190	0.3886	0.8961	0.3911	0.8961	1.1561	0.6370			
10		-0.7314	-0.1477	0.1496	0.2602	0.9429	0.5118	0.9429	0.4602	0.3819	-0.1875		
11		-0.4575	0.1448	0.5519	0.6655	0.5132	0.8016	0.5132	0.3079	0.2250	0.0601	-0.2760	
12		-0.5507	0.0206	0.4684	0.4261	0.6562	0.6562	0.3216	0.3457	0.5298	-0.2144	0.2505	-0.2011
13		-0.2900	-0.4087	-0.4816	-0.7107	-0.1165	-0.0406	-0.1985	-0.6492	0.0764	-0.0938	-0.4542	-1.2362
14	ω_1	-0.5669	-0.5373	-1.0994	-0.5933	-0.4861	0.3561	-0.3717	-0.8934	-0.7253	-1.1374	-1.2656	-1.2444
15		-1.2369	-0.9417	-0.9636	-0.0215	0.5062	0.3957	-0.4069	-0.6832	-1.0893	-1.3779		
16		-0.4537	-0.4501	0.8502	0.9048	1.1668	1.1668	0.4412	-0.4750	-0.6717			
17		-0.9227	-0.1853	0.8751	0.7398	0.5869	0.5869	-0.6992	-0.8246				
18		-0.0675	-0.3684	0.0784	-0.2742	-0.5368	-0.9467						
19		-0.8053	-0.9131	-0.6913	-1.5204	-1.0707							
20		-0.6100	-0.8914	-0.4349	-1.0976								
21		-0.7282	-0.9458	-0.8685									
22		-0.5177	-1.1897	-0.4677									
23		-0.7791	-0.9275										
24	π	-0.8179											

FIG. 5.5 $P(\omega_1, \omega_2, \omega_3)$

Based on first differences of logarithms of Standard and Poor Stock
Price Index with constant equal to minimum price plus one subtracted
from each observation
(1020 observations, 25 terms in each of four iterations of filter on
complex-demodulates)

the amount of money in the 'float' varies automatically as the demands of the banks change, we would expect a seasonal pattern in the series. While the float is a 'residual' in terms of the monetary policies of the Federal Reserve, changes in monetary policy do have effects on the float through their effects on the activities of member banks. In April of 1951 a major change occurred in monetary policy on the basis of the post-war recovery of the economy. It is interesting to see how this change effected the behaviour of the federal float series.

Thus we see in Fig. 5.6 that the complex-demodulates at some of the seasonal frequencies changed remarkably while others were very little effected. If we were attempting to predict this series, information of this kind would naturally lead us to be very cautious about the application of linear methods which relied on time averages over this period.

When we look at the bi-spectrum of this series we find considerable evidence that the seasonal frequencies come about in part from a quadratic mechanism. Fig. 5.6 shows the magnitude of the bi-spectrum, while Fig. 5.7 shows the bi-spectral coefficients. The seasonal frequencies occur at $\omega = \pi/6, \pi/3, \pi/2, 2\pi/3, 5\pi/6, \pi$. These frequencies correspond to the following ordinates of the Figures: 4, 8, 12, 16, 20, 24.

On the basis of these results we are led to reject the hypothesis that the entire process including the seasonal frequencies is well represented as a linear process. While these estimates do not permit us to, as yet, specify the exact mechanism which may be generating power in the spectrum and bi-spectrum at these frequencies, they do provide information which is suggestive of a variety of further analyses.

7. Conclusion

A very preliminary report on the application of complex-demodulation and bi-spectrum estimation has been presented. It has been indicated that complex-demodulation provides information concerning certain kinds of non-stationarity. Bi-spectrum analysis tests the completeness of the assumption of linear processes and suggests further possible analyses which may be able to determine the form of a non-linear element.

8. Acknowledgement

Dave Brillinger originally attracted my attention to this subject and provided much valuable advice. I was very fortunate to be able to present many of the preliminary results which are reported here to John Tukey. His many helpful comments and criticisms are gratefully acknowledged.

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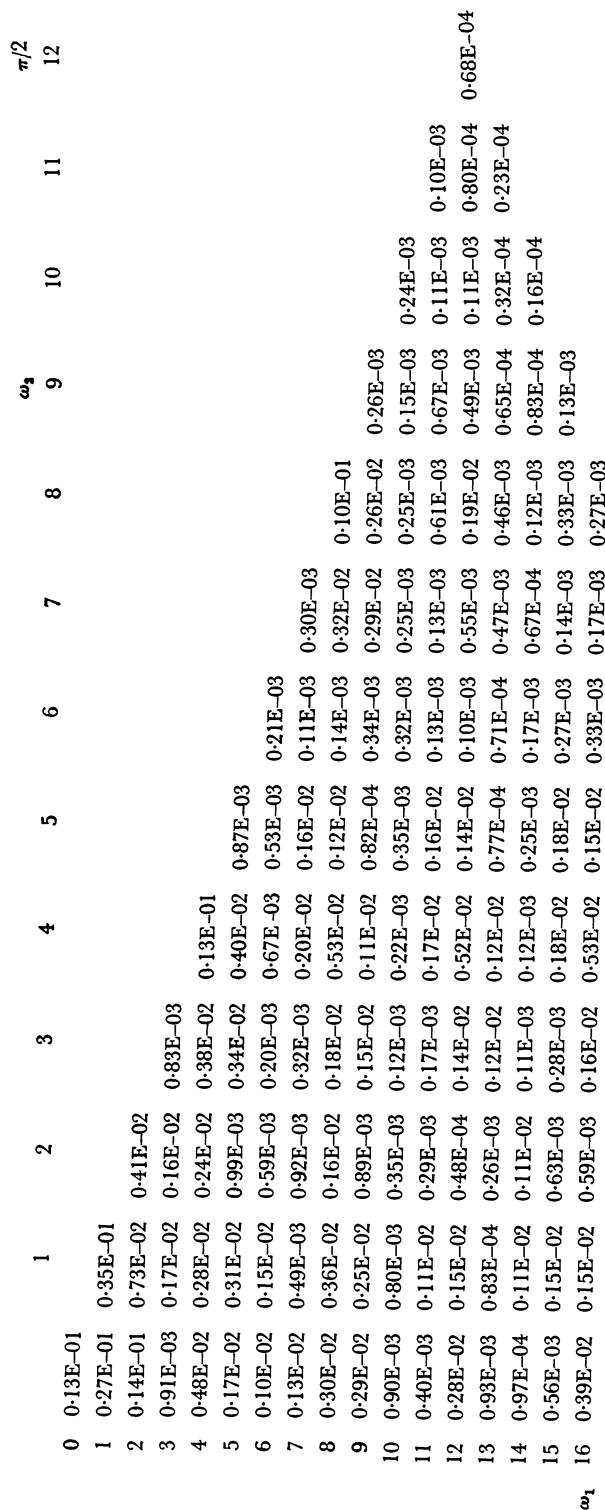


FIG. 5.6 $|f_{123}(\omega_1, \omega_2)|$
Based on logarithms of monthly Federal Float Data
227 observations
15 terms in each of four iterations of filter on complex-demodulates

	$\pi/2$											
	ω_2											
	1	2	3	4	5	6	7	8	9	10	11	12
0	0.0043											
1	0.0267	0.2546										
2	0.1598	0.1371	0.0722									
3	0.0047	0.0070	0.0875	0.2764								
4	0.0170	0.0207	0.3221	0.6633	0.8470							
5	0.0179	0.3247	0.0509	0.5135	0.7910	0.4964						
6	0.1359	0.1106	0.0264	0.0252	0.4386	0.5020	0.1658					
7	0.0220	0.0012	0.0622	0.0879	0.7838	0.7553	0.0580	0.2378				
8	0.0131	0.0671	0.2971	0.6111	0.8678	0.4786	0.0545	0.9777	1.0982			
9	0.1065	0.4404	0.1617	0.6137	0.3902	0.0127	0.8337	0.7962	0.1995			
10	0.2122	0.1236	0.0559	0.0597	0.1333	0.1325	0.0974	0.2720	0.1002	0.4458		
11	0.0160	0.0703	0.0868	0.1383	0.5533	0.0517	0.1384	0.3392	0.4860	0.1876	0.3204	
12	0.1895	0.2429	0.0021	0.5247	0.6365	0.1248	0.4057	0.6378	0.3999	0.3045	0.0961	0.0293
13	0.2824	0.0048	0.0254	0.4655	0.0617	0.0785	0.4329	0.4305	0.0841	0.0918	0.0264	
14	0.0103	0.1794	0.2827	0.0261	0.3854	0.3134	0.0512	0.1723	0.2525	0.0377		
15	0.0041	0.0110	0.0347	0.1492	0.8542	0.2668	0.0816	0.1471	0.0625			
16	0.0212	0.0134	0.0844	0.5819	0.6745	0.4374	0.0374	0.0272				
17	0.0562	0.0274	0.0204	0.6047	0.1508	0.0563	0.0251					
18	0.1919	0.0242	0.1343	0.3259	0.4698	0.1661						
19	0.0020	0.0391	0.1747	0.2492	0.1430							
20	0.0135	0.0174	0.1717	0.1124	0.0924							
21	0.0045	0.0996	0.0434	0.1925								
22	0.4767	0.0887	0.0277									
23	0.1916	0.0009										
24	0.0274											
ω_1												
π												

FIG. 5.7 $\hat{S}_3(\omega_1, \omega_2, \omega_3)$
Based on original Monthly Federal Float Data
227 observations
15 terms in each of four iterations of filter on complex-demodulates

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